$$C_{qu}^{(1)\,i1\_i2\_i3\_i4\_} \to$$

$$\frac{1}{16\,\pi^2}\ \Bigg(\frac{4}{243}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{d}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{5}{162}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{d}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{16}{1215}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{d}}{}^p\right]$$

$$\delta_{i1i2}\,\delta_{i3i4}+\frac{4}{81}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{e}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{5}{54}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{e}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{16}{405}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{e}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{l}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{5}{108}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{l}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{8}{405}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{l}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{243}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{q}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{5}{324}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{q}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{8}{1215}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{q}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{16}{243}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\left[m_{\tilde{u}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{10}{81}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\left[m_{\tilde{u}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{64}{1215}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\left[m_{\tilde{u}}{}^p\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1{}^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{27}\,g_1{}^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{16}{405}\,g_1{}^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{243}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_1,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{2,2,-1}\left[m_1,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{243}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_1,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{4,1,-2}\left[m_1,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,1,0}\left[m_2,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,2,-1}\left[m_2,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,1,-1}\left[m_2,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{4,1,-2}\left[m_2,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,1,0}\left[m_2,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,2,-1}\left[m_2,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,1,-1}\left[m_2,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{4,1,-2}\left[m_2,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde{q}}{}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde{q}}{}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_3,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,2,-1}\left[m_3,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_3,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{4,1,-2}\left[m_3,m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1{}^2\,\overline{y_d}^{i2p}\,y_d{}^{i1p}\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{d}}{}^p,\tilde{\mu}\right]\delta_{i3i4}-\frac{1}{27}\,g_1{}^2\,\overline{y_d}^{i2p}\,y_d{}^{i1p}\,\mathrm{LF}_{2,2,-1}\left[m_{\tilde{d}}{}^p,\tilde{\mu}\right]\delta_{i3i4}-$$

$$\frac{1}{27}\,g_1{}^2\,\overline{y_d}^{i2p}\,y_d{}^{i1p}\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{d}}{}^p,\tilde{\mu}\right]\delta_{i3i4}-\frac{1}{6}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{3,1,0}\left[m_{\tilde{l}}{}^p,m_{\tilde{e}}{}^r\right]+$$

$$\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}{}^p,m_{\tilde{e}}{}^r\right]-\frac{1}{3}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}{}^p,m_{\tilde{e}}{}^r\right]-$$

$$\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r\right]+\frac{3}{2}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r\right]-$$

$$\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r\right]-\frac{1}{54}\,g_1{}^2\,\overline{y_u}^{pi3}\,y_u{}^{pi4}\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}{}^p,\tilde{\mu}\right]\delta_{i1i2}+$$

$$\frac{1}{108}\,g_1{}^2\,\overline{y_u}^{pi3}\,y_u{}^{pi4}\,\mathrm{LF}_{2,2,-1}\left[m_{\tilde{q}}{}^p,\tilde{\mu}\right]\delta_{i1i2}+\frac{1}{108}\,g_1{}^2\,\overline{y_u}^{pi3}\,y_u{}^{pi4}\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}{}^p,\tilde{\mu}\right]\delta_{i1i2}-$$

$$\frac{1}{972}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}{}^{i1},m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}{}^{i1},m_1\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}{}^{i1},m_2\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}{}^{i1},m_2\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}{}^{i1},m_3\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}{}^{i1},m_3\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{972}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}{}^{i2},m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{1944}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}{}^{i2},m_1\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{36}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}{}^{i2},m_2\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{72}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}{}^{i2},m_2\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{q}}{}^{i2},m_3\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{q}}{}^{i2},m_3\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{27}\,g_1{}^2\,\overline{y_u}^{i2p}\,y_u{}^{i1p}\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{u}}{}^p,\tilde{\mu}\right]\delta_{i3i4}+\frac{2}{27}\,g_1{}^2\,\overline{y_u}^{i2p}\,y_u{}^{i1p}\,\mathrm{LF}_{2,2,-1}\left[m_{\tilde{u}}{}^p,\tilde{\mu}\right]\delta_{i3i4}+$$

$$\frac{2}{27}\,g_1{}^2\,\overline{y_u}^{i2p}\,y_u{}^{i1p}\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{u}}{}^p,\tilde{\mu}\right]\delta_{i3i4}-\frac{1}{2}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\right]+$$

$$\frac{3}{2}\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\right]-\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\right]-$$

$$\frac{4}{243}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{u}}{}^{i3},m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{u}}{}^{i3},m_1\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{u}}{}^{i3},m_3\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{u}}{}^{i3},m_3\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{243}\,g_1{}^4\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{u}}{}^{i4},m_1\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{243}\,g_1{}^4\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{u}}{}^{i4},m_1\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{2,1,0}\left[m_{\tilde{u}}{}^{i4},m_3\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{81}\,g_1{}^2\,g_3{}^2\,\mathrm{LF}_{3,1,-1}\left[m_{\tilde{u}}{}^{i4},m_3\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{3}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_1]-\frac{2}{3}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_1]+$$

$$\frac{1}{3}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_1]+g_2{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_2]-$$

$$2\,g_2{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_2]+g_2{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_2]-$$

$$\frac{1}{27}\,g_1{}^2\,\overline{y_d}^{i2p}\,y_d{}^{i1p}\,\mathrm{LF}_{2,1,0}\left[\tilde{\mu},m_{\tilde{d}}{}^p\right]\delta_{i3i4}-\frac{5}{54}\,g_1{}^2\,\overline{y_d}^{i2p}\,y_d{}^{i1p}\,\mathrm{LF}_{3,1,-1}\left[\tilde{\mu},m_{\tilde{d}}{}^p\right]\delta_{i3i4}+$$

$$\frac{1}{54}\,g_1{}^2\,\overline{y_d}^{i2p}\,y_d{}^{i1p}\,\mathrm{LF}_{4,1,-2}\left[\tilde{\mu},m_{\tilde{d}}{}^p\right]\delta_{i3i4}+\frac{1}{108}\,g_1{}^2\,\overline{y_u}^{pi3}\,y_u{}^{pi4}\,\mathrm{LF}_{2,1,0}\left[\tilde{\mu},m_{\tilde{q}}{}^p\right]\delta_{i1i2}-$$

$$\frac{11}{108}\,g_1{}^2\,\overline{y_u}^{pi3}\,y_u{}^{pi4}\,\mathrm{LF}_{3,1,-1}\left[\tilde{\mu},m_{\tilde{q}}{}^p\right]\delta_{i1i2}+\frac{1}{27}\,g_1{}^2\,\overline{y_u}^{pi3}\,y_u{}^{pi4}\,\mathrm{LF}_{4,1,-2}\left[\tilde{\mu},m_{\tilde{q}}{}^p\right]\delta_{i1i2}+$$

$$\frac{2}{27}\,g_1{}^2\,\overline{y_u}^{i2p}\,y_u{}^{i1p}\,\mathrm{LF}_{2,1,0}\left[\tilde{\mu},m_{\tilde{u}}{}^p\right]\delta_{i3i4}+\frac{1}{54}\,g_1{}^2\,\overline{y_u}^{i2p}\,y_u{}^{i1p}\,\mathrm{LF}_{3,1,-1}\left[\tilde{\mu},m_{\tilde{u}}{}^p\right]\delta_{i3i4}+$$

$$\frac{1}{54}\,g_1{}^2\,\overline{y_u}^{i2p}\,y_u{}^{i1p}\,\mathrm{LF}_{4,1,-2}\left[\tilde{\mu},m_{\tilde{u}}{}^p\right]\delta_{i3i4}-\frac{1}{162}\,g_1{}^4\,\mathrm{LF}_{2,1,1,-1}\left[m_1,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{81}\,g_1{}^4\,m_1{}^2\,\mathrm{LF}_{2,1,1,0}\left[m_1,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{108}\,m_1\,g_1{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}$$

$$\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i4}\right]+\frac{1}{108}\,m_1\,g_1{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i3}\right]-$$

$$\frac{1}{162}\,g_1{}^4\,\mathrm{LF}_{2,1,1,-1}\left[m_1,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{81}\,g_1{}^4\,m_1{}^2\,\mathrm{LF}_{2,1,1,0}\left[m_1,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{108}\,m_1\,g_1{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde{u}}{}^{i3},m_{\tilde{q}}{}^{i2}\right]+$$

$$\frac{1}{108}\,m_1\,g_1{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-1}\left[m_1,m_{\tilde{u}}{}^{i4},m_{\tilde{q}}{}^{i1}\right]+\frac{1}{72}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde{\mu},m_{\tilde{q}}{}^{i1}\right]+$$

$$\frac{1}{72}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde{\mu},m_{\tilde{q}}{}^{i2}\right]-\frac{1}{18}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde{\mu},m_{\tilde{u}}{}^{i3}\right]-$$

$$\frac{1}{18}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-2}\left[m_1,\tilde{\mu},m_{\tilde{u}}{}^{i4}\right]-\frac{1}{8}\,g_2{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-2}\left[m_2,\tilde{\mu},m_{\tilde{q}}{}^{i1}\right]-$$

$$\frac{1}{8}\,g_2{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-2}\left[m_2,\tilde{\mu},m_{\tilde{q}}{}^{i2}\right]-\frac{1}{9}\,g_3{}^4\,\mathrm{LF}_{2,1,1,-1}\left[m_3,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}\,g_3{}^4\,m_3{}^2\,\mathrm{LF}_{2,1,1,0}\left[m_3,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}\,m_3\,g_3{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i4}\right]+$$

$$\frac{1}{9}\,m_3\,g_3{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i3}\right]-\frac{1}{9}\,g_3{}^4\,\mathrm{LF}_{2,1,1,-1}\left[m_3,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{9}\,g_3{}^4\,m_3{}^2\,\mathrm{LF}_{2,1,1,0}\left[m_3,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{9}\,m_3\,g_3{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde{u}}{}^{i3},m_{\tilde{q}}{}^{i2}\right]+$$

$$\frac{1}{9}\,m_3\,g_3{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-1}\left[m_3,m_{\tilde{u}}{}^{i4},m_{\tilde{q}}{}^{i1}\right]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{i2p}\,y_d{}^{rp}\,\overline{y_u}^{ri3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{d}}{}^p,m_{\tilde{q}}{}^r,\tilde{\mu}\right]+\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{i2p}\,y_d{}^{rp}\,\overline{y_u}^{ri3}\,y_u{}^{i1i4}$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{d}}{}^p,m_{\tilde{q}}{}^r,\tilde{\mu}\right]+\frac{1}{24}\,\tilde{\mu}^2\,\overline{y_d}^{i2p}\,y_d{}^{rp}\,\overline{y_u}^{ri3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{d}}{}^p,\tilde{\mu},m_{\tilde{q}}{}^r\right]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,\overline{y_u}^{i2i3}\,y_u{}^{pi4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{d}}{}^r,m_{\tilde{q}}{}^p,\tilde{\mu}\right]+\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,\overline{y_u}^{i2i3}\,y_u{}^{pi4}$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{d}}{}^r,m_{\tilde{q}}{}^p,\tilde{\mu}\right]+\frac{1}{24}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,\overline{y_u}^{i2i3}\,y_u{}^{pi4}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{d}}{}^r,\tilde{\mu},m_{\tilde{q}}{}^p\right]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,\overline{y_u}^{i2i3}\,y_u{}^{pi4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r,\tilde{\mu}\right]+\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,\overline{y_u}^{i2i3}\,y_u{}^{pi4}$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r,\tilde{\mu}\right]+\frac{1}{24}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{i1r}\,\overline{y_u}^{i2i3}\,y_u{}^{pi4}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}{}^p,\tilde{\mu},m_{\tilde{d}}{}^r\right]-$$

$$\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{i2p}\,y_d{}^{rp}\,\overline{y_u}^{ri3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}{}^r,m_{\tilde{d}}{}^p,\tilde{\mu}\right]+\frac{1}{12}\,\tilde{\mu}^2\,\overline{y_d}^{i2p}\,y_d{}^{rp}\,\overline{y_u}^{ri3}\,y_u{}^{i1i4}$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}{}^r,m_{\tilde{d}}{}^p,\tilde{\mu}\right]+\frac{1}{24}\,\tilde{\mu}^2\,\overline{y_d}^{i2p}\,y_d{}^{rp}\,\overline{y_u}^{ri3}\,y_u{}^{i1i4}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}{}^r,\tilde{\mu},m_{\tilde{d}}{}^p\right]-$$

$$\frac{1}{54}\,m_1\,g_1{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}{}^{i1},m_1,m_{\tilde{u}}{}^{i4}\right]+\frac{1}{54}\,m_1\,g_1{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}{}^{i1},m_1,m_{\tilde{u}}{}^{i4}\right]-\frac{2}{9}\,m_3\,g_3{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}{}^{i1},m_3,m_{\tilde{u}}{}^{i4}\right]+$$

$$\frac{2}{9}\,m_3\,g_3{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}{}^{i1},m_3,m_{\tilde{u}}{}^{i4}\right]-\frac{1}{54}\,m_1\,g_1{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}{}^{i2},m_1,m_{\tilde{u}}{}^{i3}\right]+\frac{1}{54}\,m_1\,g_1{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}{}^{i2},m_1,m_{\tilde{u}}{}^{i3}\right]-$$

$$\frac{2}{9}\,m_3\,g_3{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}{}^{i2},m_3,m_{\tilde{u}}{}^{i3}\right]+\frac{2}{9}\,m_3\,g_3{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}{}^{i2},m_3,m_{\tilde{u}}{}^{i3}\right]-\frac{1}{54}\,m_1\,g_1{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}{}^{i3},m_1,m_{\tilde{q}}{}^{i2}\right]+$$

$$\frac{1}{54}\,m_1\,g_1{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}{}^{i3},m_1,m_{\tilde{q}}{}^{i2}\right]-\frac{2}{9}\,m_3\,g_3{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}$$

$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}{}^{i3},m_3,m_{\tilde{q}}{}^{i2}\right]+\frac{2}{9}\,m_3\,g_3{}^2\,y_u{}^{i1i4}\,\overline{a_u}^{i2i3}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}{}^{i3},m_3,m_{\tilde{q}}{}^{i2}\right]-$$

$$\frac{1}{54}\,m_1\,g_1{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}{}^{i4},m_1,m_{\tilde{q}}{}^{i1}\right]+\frac{1}{54}\,m_1\,g_1{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}$$

$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}{}^{i4},m_1,m_{\tilde{q}}{}^{i1}\right]-\frac{2}{9}\,m_3\,g_3{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}{}^{i4},m_3,m_{\tilde{q}}{}^{i1}\right]+$$

$$\frac{2}{9}\,m_3\,g_3{}^2\,\overline{y_u}^{i2i3}\,a_u{}^{i1i4}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}{}^{i4},m_3,m_{\tilde{q}}{}^{i1}\right]-\frac{1}{2}\,\tilde{\mu}^2\,\overline{y_d}^{i2p}\,y_d{}^{i1p}\,\overline{y_u}^{ri3}\,y_u{}^{ri4}$$

$$\mathrm{LF}_{2,1,1,0}\left[\tilde{\mu},m_{\tilde{d}}{}^p,m_{\tilde{q}}{}^r\right]-\frac{1}{4}\,\overline{y_u}^{ri3}\,\overline{y_u}^{i2p}\,y_u{}^{ri4}\,y_u{}^{i1p}\,\mathrm{LF}_{2,1,1,-1}\left[\tilde{\mu},m_{\tilde{q}}{}^r,m_{\tilde{u}}{}^p\right]+$$

$$\frac{1}{216}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde{q}}{}^{i1},m_{\tilde{q}}{}^{i2},\tilde{\mu}\right]-\frac{1}{54}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i3},\tilde{\mu}\right]-\frac{1}{54}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i4},\tilde{\mu}\right]+$$

$$\frac{2}{27}\,g_1{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}\left[m_1,m_{\tilde{u}}{}^{i3},m_{\tilde{u}}{}^{i4},\tilde{\mu}\right]+\frac{1}{8}\,g_2{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}\left[m_2,m_{\tilde{q}}{}^{i1},m_{\tilde{q}}{}^{i2},\tilde{\mu}\right]+\frac{2}{9}\,g_3{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde{q}}{}^{i1},m_{\tilde{q}}{}^{i2},\tilde{\mu}\right]-$$

$$\frac{2}{9}\,g_3{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde{q}}{}^{i1},m_{\tilde{u}}{}^{i3},\tilde{\mu}\right]-\frac{2}{9}\,g_3{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}$$

$$\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde{q}}{}^{i2},m_{\tilde{u}}{}^{i4},\tilde{\mu}\right]+\frac{2}{9}\,g_3{}^2\,\overline{y_u}^{i2i3}\,y_u{}^{i1i4}\,\mathrm{LF}_{1,1,1,1,-1}\left[m_3,m_{\tilde{u}}{}^{i3},m_{\tilde{u}}{}^{i4},\tilde{\mu}\right]\Bigg)$$